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## Supplemental Information

# Cohort studies of long-term exposure to outdoor particulate matter and risks of cancer: A systematic review and meta-analysis

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**Supplementary materials**

**Table S1:** Searching strategy

**Table S2:** Quality Assessment Tool for Observational Cohort and Cross-Sectional Studies

**Table S3:** Quality Assessment for included studies

**Table S4:** GRADE assessment for the association between exposure to $PM_{2.5}$ and risk of lung cancer

**Table S5:** GRADE assessment for the association between exposure to $PM_{10}$ and risk of lung cancer

**Table S6:** GRADE assessment for the association between exposure to $PM_{2.5}$and risk of breast cancer incidence

**Table S7:** GRADE assessment for the association between exposure to $PM_{2.5}$and risk of breast cancer mortality

**Table S8:** GRADE assessment for the association between exposure to $PM_{2.5}$ and risk of breast cancer incidence

**Table S9:** GRADE assessment for the association between exposure to PM and risk of other cancers

**Table S10**: Sensitivity analysis of lung cancer associated a 10-μg/$m^3$ change in exposure to $PM_{2.5}$ and $PM_{10}$ sorted by $I^2$

**Table S11.** Associations between $PM_{2.5}$ and $PM_{10}$ and risk for lung cancer, according to WHO air pollution threshold.

**Figure S1**: Estimates of lung cancer risk associated with a 10-μg/$m^3$ change in exposure to $PM_{2.5}$ or $PM_{10}$ overall and by region

**Figure S2**: Funnel Plots with trim and fill analysis of studies about $PM_{2.5}$ (A) or $PM_{10}$ (B) and lung cancer risk.

**Figure S3**: Funnel Plots with trim and fill analysis of studies about $PM_{2.5}$ (*A for incidence, B for mortality*) or $PM_{10}$ (*C for incidence*) and breast cancer risk

**Table S1:** Searching strategy

| | |
|---|---|
| #1 | (Neoplas* or tumor* or cancer* or Melano* or leukem* or leaukaem* or lymph* or adenocarcinoma* or hemangioma*).mp. |
| #2 | (Particul* adj3 matter*) or (Particul* and air pollut*) or PM 2 adj1 5 or PM 10 or PM 1).mp. |
| #3 | (cohort* or follow-up or follow* or ((inciden* or concurren* or prospect* or retrospect* or Longitu*) and Stud*)).mp. |
| #4 | exp animals/ not humans.sh. |
| #5 | #1 AND #2 AND #3 |
| #6 | #5 NOT #4 |

**Table S2:** Quality Assessment Tool for Observational Cohort and Cross-Sectional Studies.

| Criteria | |
|---|---|
| 1. Was the research question or objective in this paper clearly stated? | ○ |
| 2. Was the study population clearly specified and defined? | ○ |
| 3. Was the participation rate of eligible persons at least 50%? | |
| 4. Were all the subjects selected or recruited from the same or similar populations (including the same time period)? Were inclusion and exclusion criteria for being in the study prespecified and applied uniformly to all participants? | ○ |
| 5. Was a sample size justification, power description, or variance and effect estimates provided? | |
| 6. For the analyses in this paper, were the exposure(s) of interest measured prior to the outcome(s) being measured? | ○ |
| 7. Was the timeframe sufficient so that one could reasonably expect to see an association between exposure and outcome if it existed? | ○ |
| 8. For exposures that can vary in amount or level, did the study examine different levels of the exposure as related to the outcome (e.g., categories of exposure, or exposure measured as continuous variable)? | ○ |
| 9. Were the exposure measures (independent variables) clearly defined, valid, reliable, and implemented consistently across all study participants? | ○ |
| 10. Was the exposure(s) assessed more than once over time? | |
| 11. Were the outcome measures (dependent variables) clearly defined, valid, reliable, and implemented consistently across all study participants? | ○ |
| 12. Were the outcome assessors blinded to the exposure status of participants? | |
| 13. Was loss to follow-up after baseline 20% or less? | |
| 14. Were key potential confounding variables measured and adjusted statistically for their impact on the relationship between exposure(s) and outcome(s)? | ○ |

(Source: https://www.nhlbi.nih.gov/health-topics/study-quality-assessment-tools)
Note: Criterias with ○ were included in the quality assessment in this study.

**Table S3:** Quality Assessment for included studies

| No. | Reference | Score |
|---|---|---|
| 1 | Coleman et al. 2020 | Good |
| 2 | Guo et al. 2020 | Good |
| 3 | Bai et al. 2019 | Good |
| 4 | Cheng et al. 2019 | Fair |
| 5 | DuPre et al. 2019 | Good |
| 6 | Pope et al. 2019 | Good |
| 7 | White et al. 2019 | Good |
| 8 | Yorifuji et al. 2019 | Good |
| 9 | Andersen et al. 2018 | Good |
| 10 | Cakmak et al. 2018 | Good |
| 11 | Datzmann et al. 2018 | Fair |
| 12 | Gandini et al. 2018 | Good |
| 13 | Nagel et al. 2018 | Good |
| 14 | Pedersen et al. 2018 | Good |
| 15 | Villeneuve et al. 2018 | Good |
| 16 | Andersen et al. 2017 | Good |
| 17 | Gharibvand et al. 2017 | Fair |
| 18 | Gharibvand et al. 2017‡ | Fair |
| 19 | Pedersen et al. 2017 | Good |
| 20 | Pun et al. 2017 | Fair |
| 21 | Turner et al. 2017 | Good |
| 22 | Yin et al. 2017 | Good |
| 23 | Chen et al. 2016 | Good |
| 24 | Hart et al. 2016 | Good |
| 25 | Jorgensen et al. 2016 | Fair |
| 26 | Raaschou et al. 2016 | Good |
| 27 | Tomczak et al. 2016 | Good |
| 28 | Wong et al. 2016 | Good |
| 29 | Fischer et al. 2015 | Fair |
| 30 | Hart et al. 2015 | Good |
| 31 | To et al. 2015 | Good |
| 32 | Turner et al. 2015 | Good |
| 33 | Puett et al. 2014 | Fair |
| 34 | Carey et al. 2013 | Fair |
| 35 | Cesaroni et al. 2013 | Good |
| 36 | Heinrich et al. 2013 | Fair |
| 37 | Raaschou et al. 2013 | Good |

| | | |
|---|---|---|
| 38 | Hales et al. 2012 | Fair |
| 39 | Lepeule et al. 2012 | Good |
| 40 | Hart et al. 2011 | Fair |
| 41 | Katanoda et al. 2011 | Fair |
| 42 | Lipsett et al. 2011 | Fair |
| 43 | Turner et al. 2011ǂ | Good |
| 44 | Brunekreef et al. 2009 | Good |
| 45 | Pope et al. 2002 | Good |
| 46 | Abbey et al. 1999 | Good |
| 47 | Beeson et al. 1998 | Fair |

Notes: Sorted by year of publication and author's name; ǂ means excluded in full analysis but included in subgroup analysis

**Table S4:** GRADE assessment for the association between exposure to $PM_{2.5}$ and risk of lung cancer

| **Domains** | **Assessment** | **Downgrading/ upgrading** |
|---|---|---|
| **Start level** | **All cohort studies** | **High** |
| Risk of bias | The overall risk of bias in all cohorts was low. | No change |
| Imprecision | All studies included reported the 95% confidence interval around the best estimate of the absolute effect. | No change |
| Inconsistency | The values of effect sizes across the studies were inconsistent. The point estimates were in the range of 0.95 to 1.63. | Downgrade |
| Indirectness | The exposure assessment methods various in studies included. | No change |
| Publication bias | Given the comprehensive search, it seems little publication bias. | No change |
| Magnitude of associations | The magnitude of effect sizes was not large enough to upgrade the level of evidence. | No change |
| Dose-response trend | No report | No change |
| Residual confounding | The risk of exposure misclassification is low in all included studies. | No change |
| Overall judgment | **Moderate** | |

**Table S5:** GRADE assessment for the association between exposure to $PM_{10}$ and risk of lung cancer

| **Domains** | **Assessment** | **Downgrading/ upgrading** |
|---|---|---|
| **Start level** | **All cohort studies** | **High** |
| Risk of bias | The overall risk of bias in all cohorts was low. | No change |
| Imprecision | All studies included reported the 95% confidence interval around the best estimate of the absolute effect. | No change |
| Inconsistency | The values of effect sizes across the studies were inconsistent. The point estimates were in the range of 0.93 to 2.39 for $PM_{10}$. | Downgrade |
| Indirectness | The exposure assessment methods various in studies included. | No change |
| Publication bias | Given the comprehensive search, it seems little publication bias. | No change |
| Magnitude of associations | The magnitude of effect sizes was not large enough to upgrade the level of evidence. | No change |
| Dose-response trend | No report | No change |
| Residual confounding | The risk of exposure misclassification is low in all included studies. | No change |
| Overall judgment | **Moderate** | |

**Table S6:** GRADE assessment for the association between exposure to $PM_{2.5}$ and risk of breast cancer incidence

| **Domains** | **Assessment** | **Downgrading/ upgrading** |
|---|---|---|
| **Start level** | **All cohort studies** | **High** |
| Risk of bias | The overall risk of bias in all cohorts was low. | Downgrade |
| Imprecision | All studies included reported the 95% confidence interval around the best estimate of the absolute effect. | No change |
| Inconsistency | The values of effect sizes across the studies were inconsistent. The point estimates were in the range of 0.89 to 1.24 for $PM_{2.5}$ on breast cancer incidence. | Downgrade |
| Indirectness | The exposure assessment methods various in studies included. | No change |
| Publication bias | Given the comprehensive search, it seems little publication bias. | No change |
| Magnitude of associations | The magnitude of effect sizes was not large enough to upgrade the level of evidence. | No change |
| Dose-response trend | No report | No change |
| Residual confounding | The risk of exposure misclassification is low in all included studies. | No change |
| Overall judgment | **Low** | |

**Table S7:** GRADE assessment for the association between exposure to $PM_{2.5}$ and risk of breast cancer mortality

| **Domains** | **Assessment** | **Downgrading/ upgrading** |
|---|---|---|
| **Start level** | **All cohort studies** | **High** |
| Risk of bias | The overall risk of bias in all cohorts was low. | Downgrade |
| Imprecision | All studies included reported the 95% confidence interval around the best estimate of the absolute effect. | No change |
| Inconsistency | The values of effect sizes across the studies were consistent though the point estimates were different. | No change |
| Indirectness | The exposure assessment methods various in studies included. | No change |
| Publication bias | Given the comprehensive search, it seems little publication bias. | No change |
| Magnitude of associations | The magnitude of effect sizes was not large enough to upgrade the level of evidence. | No change |
| Dose-response trend | No report | No change |
| Residual confounding | The risk of exposure misclassification is low in all included studies. | No change |
| Overall judgment | **Moderate** | |

**Table S8:** GRADE assessment for the association between exposure to $PM_{10}$ and risk of breast cancer incidence

| **Domains** | **Assessment** | **Downgrading/ upgrading** |
|---|---|---|
| **Start level** | **All cohort studies** | **High** |
| Risk of bias | The overall risk of bias in all cohorts was low. | Downgrade |
| Imprecision | All studies included reported the 95% confidence interval around the best estimate of the absolute effect. | No change |
| Inconsistency | The values of effect sizes across the studies were consistent though the point estimates were different. | No change |
| Indirectness | The exposure assessment methods various in studies included. | No change |
| Publication bias | Given the comprehensive search, it seems little publication bias. | No change |
| Magnitude of associations | The magnitude of effect sizes was not large enough to upgrade the level of evidence. | No change |
| Dose-response trend | No report | No change |
| Residual confounding | The risk of exposure misclassification is low in all included studies. | No change |
| Overall judgment | **Moderate** | |

**Table S9:** GRADE assessment for the association between exposure to PM and risk of other cancers

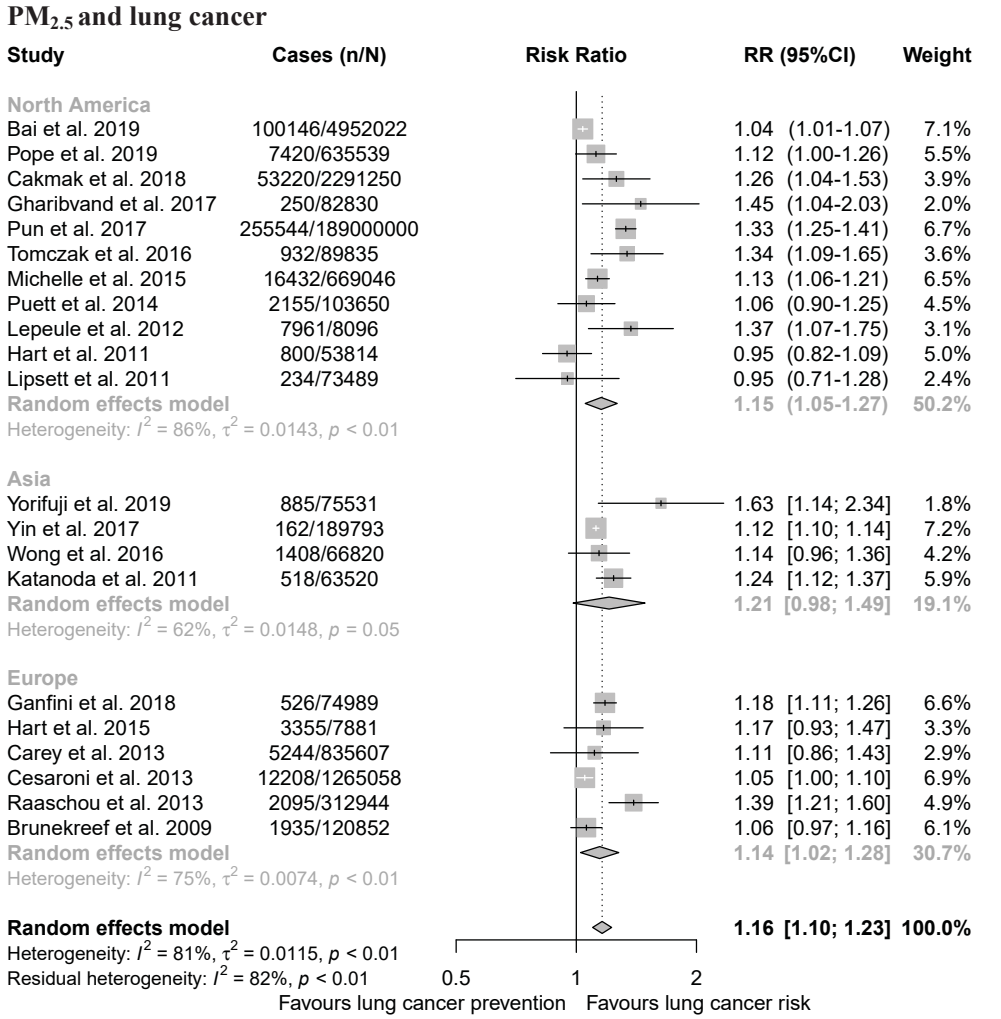
| **Domains** | **Assessment** | **Downgrading/ upgrading** |
|---|---|---|
| **Start level** | **All cohort studies** | **High** |
| Risk of bias | The overall risk of bias in all cohorts was low. | No change |
| Imprecision | All studies included reported the 95% confidence interval around the best estimate of the absolute effect. | No change |
| Inconsistency | The values of effect sizes across the studies were inconsistent. | Downgrade |
| Indirectness | The exposure assessment methods various in studies included. | No change |
| Publication bias | Given the comprehensive search, it seems little publication bias. | No change |
| Magnitude of associations | The magnitude of effect sizes was not large enough to upgrade the level of evidence. | No change |
| Dose-response trend | No report | No change |
| Residual confounding | The risk of exposure misclassification is low in all included studies. | No change |
| Overall judgment | **Moderate** | |

**Table S10.** Sensitivity analysis of lung cancer associated a 10-μg/m$^3$ change in exposure to $PM_{2.5}$ and $PM_{10}$ sorted by $I^2$

| Omitted studies | $I^2$ | Pooled relative risk (95% CI) |
|---|---|---|
| **$PM_{2.5}$** | | |
| Pun et al. 2017 | 0.72 | 1.15 (1.09-1.21) |
| Bai et al. 2019 | 0.75 | 1.17 (1.11-1.24) |
| Raaschou et al. 2013 | 0.80 | 1.15 (1.09-1.21) |
| Cesaroni et al. 2013 | 0.81 | 1.17 (1.10-1.24) |
| Hart et al. 2011 | 0.81 | 1.17 (1.11-1.23) |
| Katanoda et al. 2011 | 0.81 | 1.16 (1.09-1.23) |
| Yorifuji et al. 2019 | 0.81 | 1.15 (1.09-1.21) |
| Gandin et al. 2018 | 0.81 | 1.16 (1.09-1.23) |
| Tomczak et al. 2016 | 0.81 | 1.15 (1.09-1.22) |
| Lepeule et al. 2012 | 0.81 | 1.15 (1.09-1.22) |
| Gharibvand et al. 2017 | 0.81 | 1.15 (1.09-1.22) |
| Cakmak et al. 2018 | 0.82 | 1.16 (1.09-1.23) |
| Yin et al. 2017 | 0.82 | 1.16 (1.10-1.23) |
| Brunekreef et al. 2009 | 0.82 | 1.17 (1.10-1.24) |
| Lipsett et al. 2011 | 0.82 | 1.17 (1.10-1.23) |
| Puett et al. 2014 | 0.82 | 1.17 (1.10-1.23) |
| Turner et al. 2016 | 0.82 | 1.16 (1.10-1.23) |
| Hart et al. 2015 | 0.82 | 1.16 (1.10-1.23) |
| Wong et al. 2016 | 0.82 | 1.16 (1.10-1.23) |
| Pope et al. 2019 | 0.82 | 1.16 (1.10-1.23) |
| Carey et al. 2013 | 0.82 | 1.16 (1.10-1.23) |
| **$PM_{10}$** | | |
| Fischer et al. 2015 | 0.93 | 1.22 (1.00-1.48) |
| Heinrich et al. 2013 | 0.95 | 1.12 (0.99-1.27) |
| Hales et al. 2012 | 0.96 | 1.24 (1.02-1.50) |
| Pope et al. 2002 | 0.96 | 1.24 (1.03-1.51) |
| Beeson et al. 1998 | 0.96 | 1.18 (0.99-1.40) |
| Abbey et al. 1999(female) | 0.96 | 1.19 (0.99-1.44) |
| Raaschou et al. 2013 | 0.96 | 1.22 (1.00-1.48) |
| Lipsett et al. 2011 | 0.96 | 1.25 (1.04-1.51) |
| Chen et al. 2016 | 0.96 | 1.24 (1.02-1.50) |
| Hart et al. 2011 | 0.97 | 1.24 (1.03-1.50) |
| Puett et al. 2014 | 0.97 | 1.22 (1.00-1.48) |
| Abbey et al. 1999(male) | 0.97 | 1.22 (1.01-1.48) |
| Carey et al. 2013 | 0.97 | 1.24 (1.02-1.50) |

**Table S11.** Associations between $PM_{2.5}$ and $PM_{10}$ and risk for lung cancer, according to WHO air pollution threshold.

| | RR (95%CI) | P-value for difference |
|---|---|---|
| $PM_{2.5}$ (10-µg/$m^3$) | | |
| Lower | 1.15 (1.09-1.22) | 0.173 |
| Higher | 1.34 (1.09-1.65) | |
| $PM_{10}$ (20-µg/$m^3$) | | |
| Lower | 1.02 (1.00-1.03) | 0.019 |
| Higher | 1.27 (1.03-1.56) | |



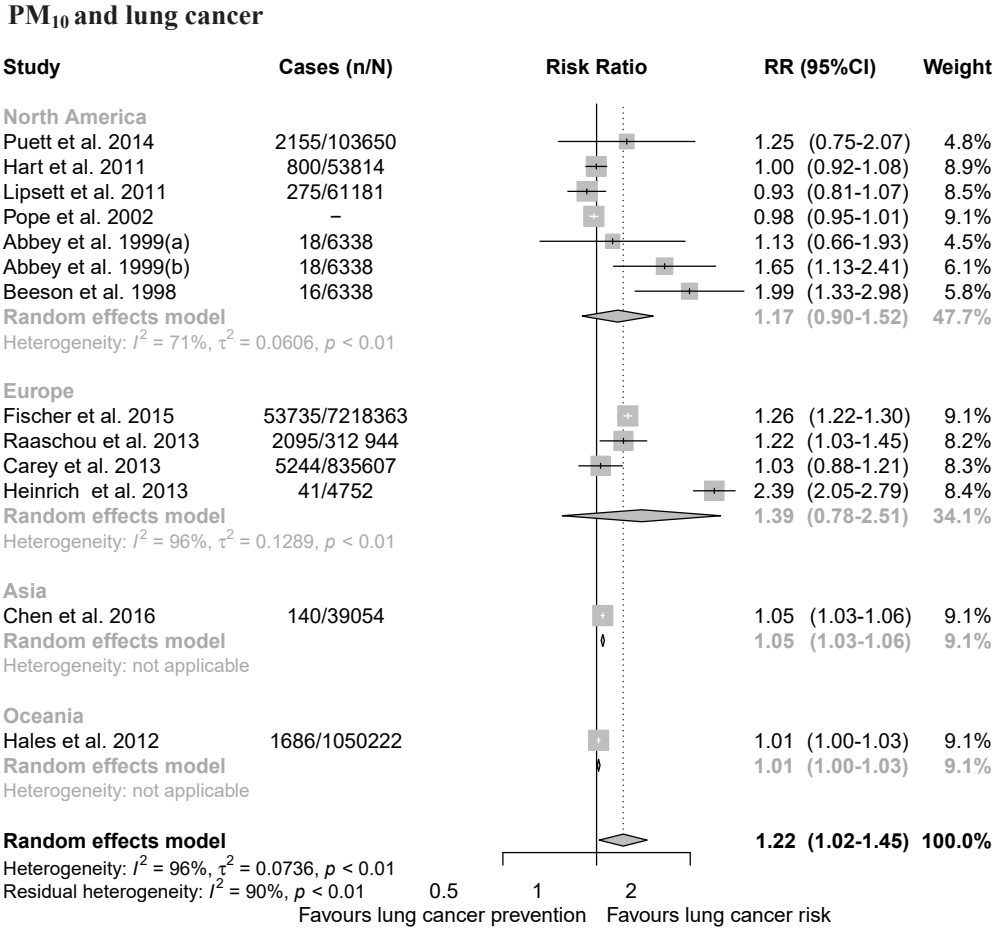
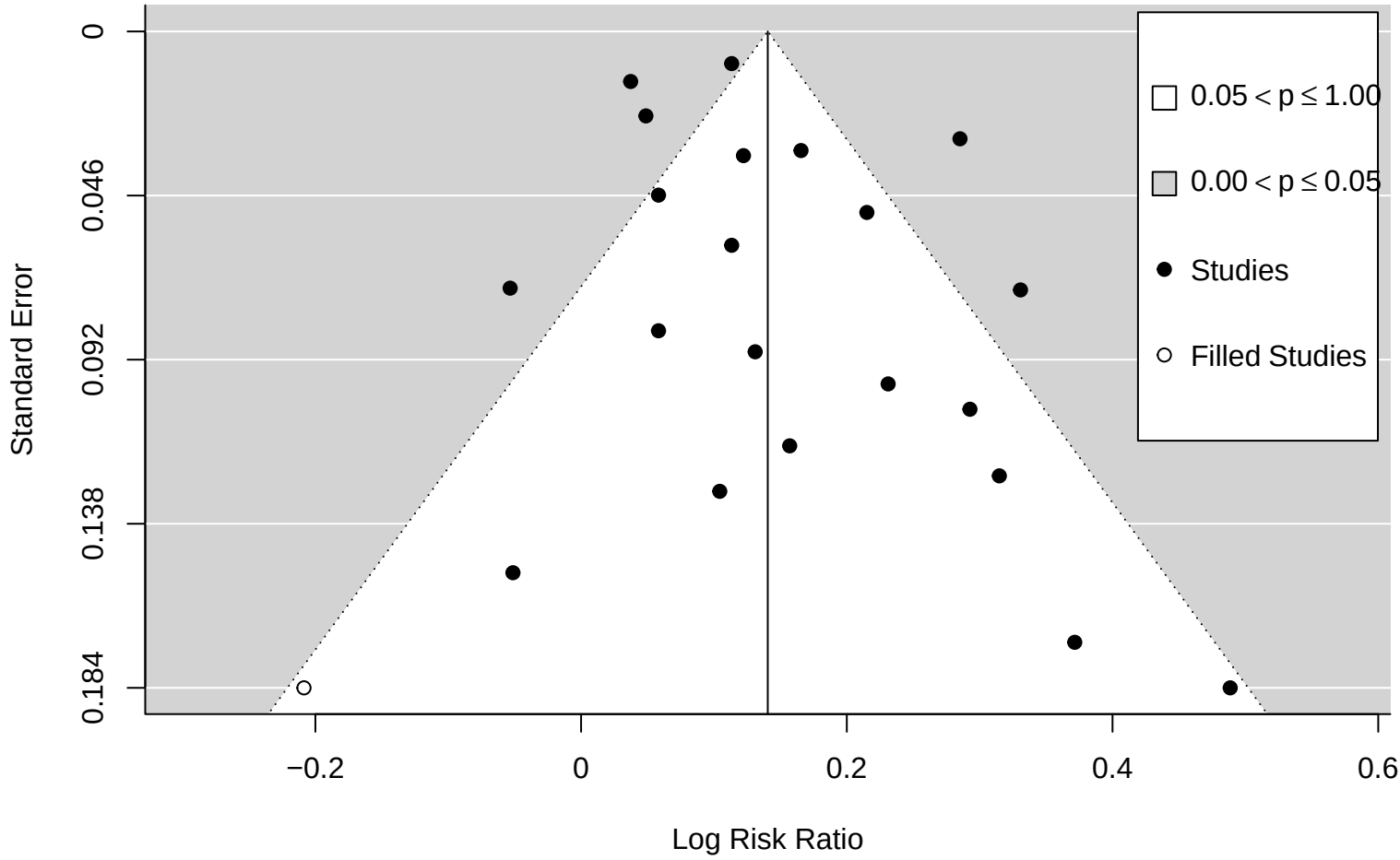




**Figure S1**: Estimates of lung cancer risk associated with a 10-μg/m$^3$ change in exposure to $PM_{2.5}$ or $PM_{10}$ overall and by region. RR= relative risk.



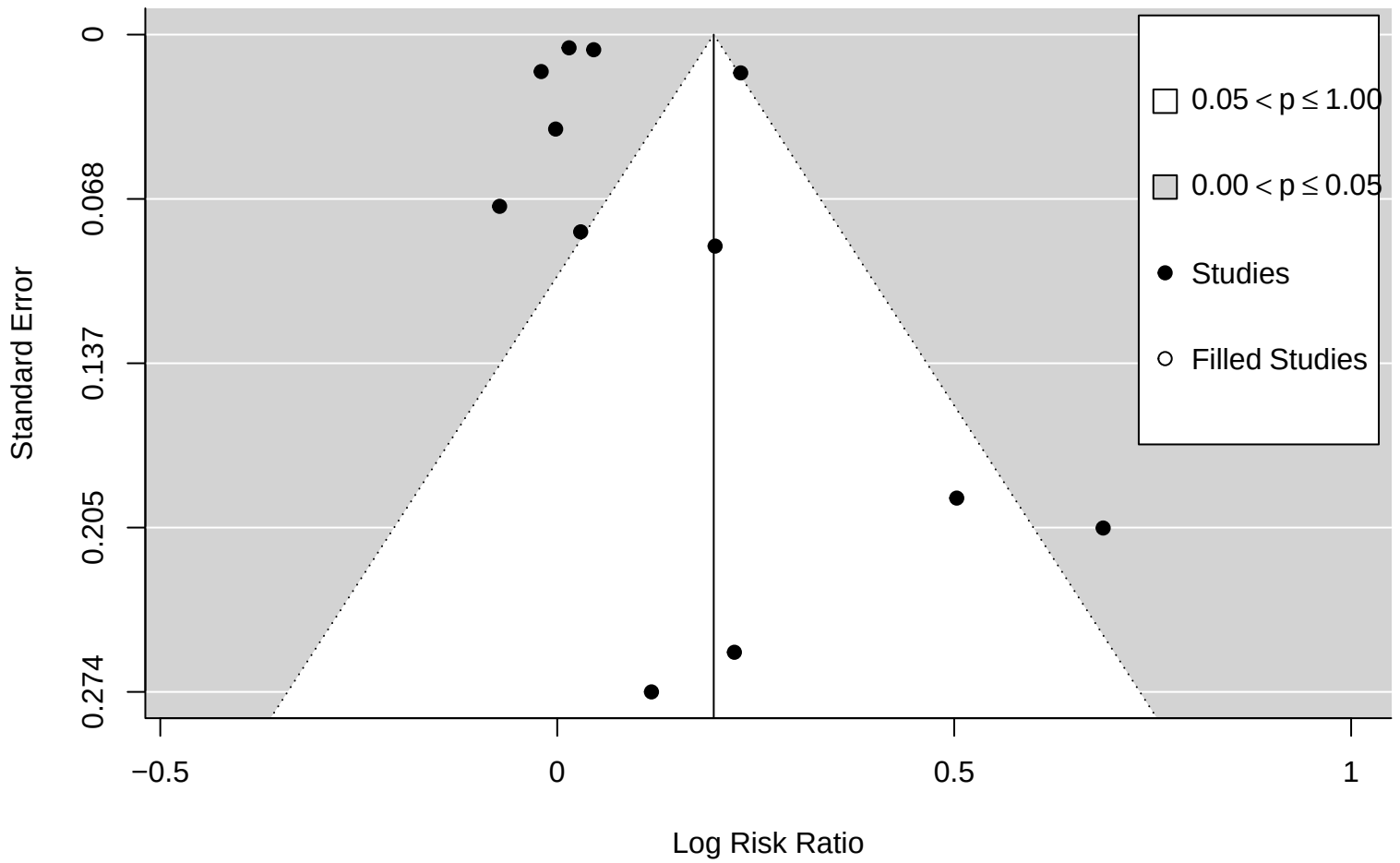
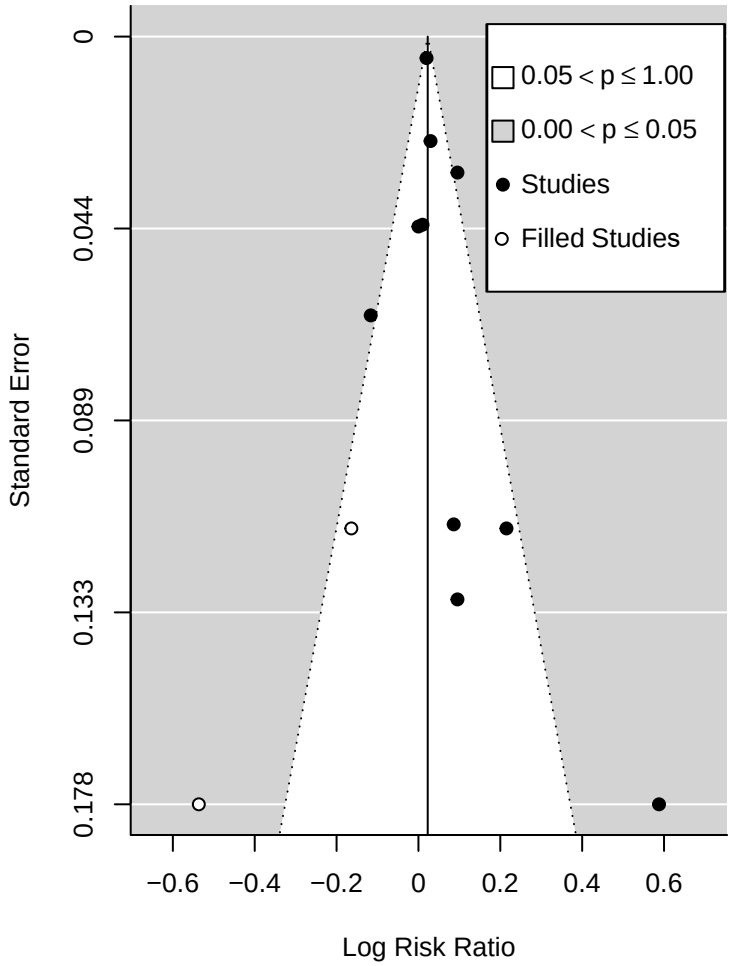
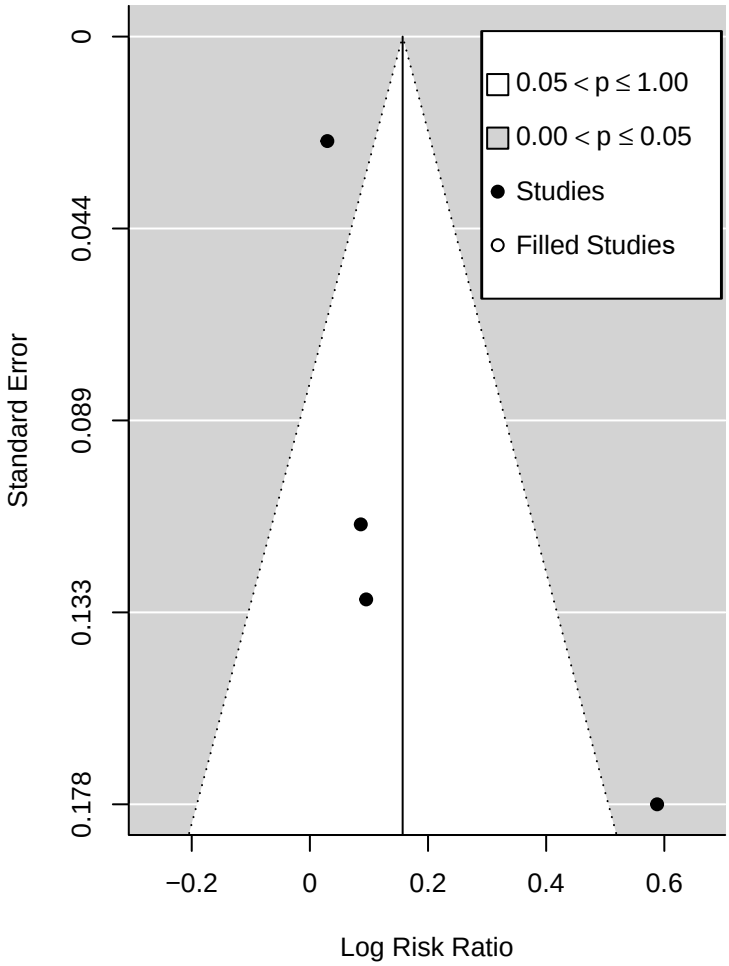




**Figure S2**: Funnel Plots with trim and fill analysis of studies about $PM_{2.5}$ (A) or $PM_{10}$ (B) and lung cancer risk.



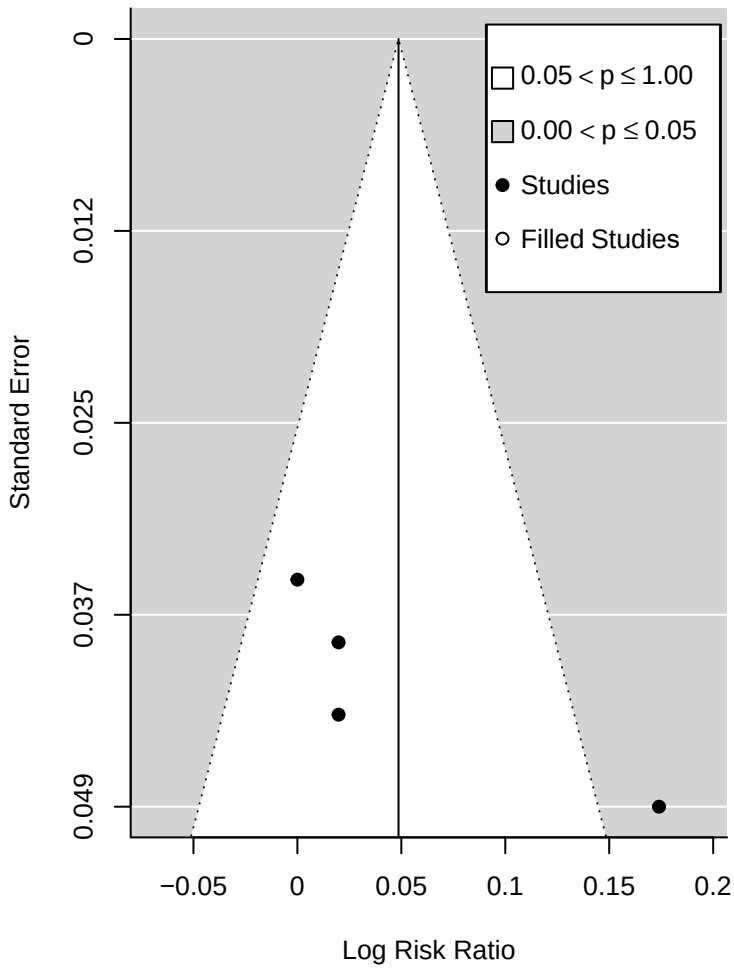




**Figure S3**: Funnel Plots with trim and fill analysis of studies about $PM_{2.5}$ (A for incidence, B for mortality) or $PM_{10}$ (C for incidence) and breast cancer risk.